

Components of an Image Processing System

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Evolution of Image Processing Systems

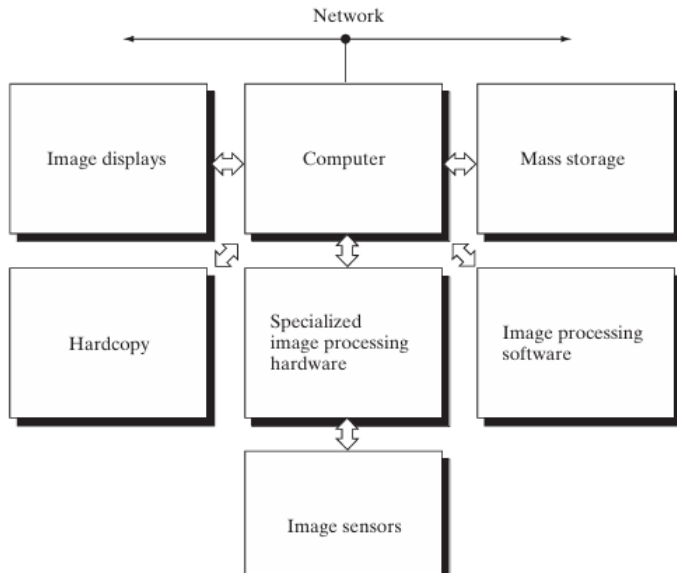
Historical Perspective

- **1980s:** Image processing systems were large peripherals attached to host computers.
- **1990s:** Shift to plug-in boards in PCs and workstations.

Current Trends

- Integration of general-purpose computing with specialized image hardware.
- Rise of image processing software companies.
- Miniaturization and increased accessibility.

Components



System Overview

General-Purpose Image Processing System Includes:

- Image sensors
- Specialized image processing hardware
- Computer
- Software
- Mass storage
- Image displays and hardcopy devices
- Network interface

Image Sensors and Digitizers

Two Main Components

- **Sensor:** Captures physical energy (e.g., light).
- **Digitizer:** Converts analog signal to digital format.

Example

In a digital camera:

- Sensor outputs voltage proportional to light intensity.
- Digitizer samples and quantizes it.

Specialized Image Processing Hardware

Purpose

Performs fast, low-level operations like:

- Arithmetic and logic processing (using ALUs)
- Noise reduction (e.g., real-time averaging)

Feature

Known as the **front-end subsystem**, it excels at high-speed data throughput, e.g., 30 fps video.

The Computer

Role

Acts as the central control and processing unit.

Types

- General-purpose PC (suitable for off-line tasks)
- High-performance systems for real-time or massive processing
- Occasionally, custom-built systems for dedicated applications

Image Processing Software

Functions

- Provides modules for filtering, enhancement, compression, etc.
- Allows scripting or programming using built-in APIs or languages.

Advanced Features

Some software allows integration with general-purpose programming (e.g., Python, MATLAB).

Mass Storage

Need for Storage

An uncompressed 1024×1024 image with 8-bit intensity needs 1 MB.

Storage Types

- 1 Short-term (RAM, frame buffers)
- 2 On-line (magnetic/optical disks)
- 3 Archival (tapes, disk jukeboxes)

Frame Buffers

Definition

Special boards that store complete images for rapid access.

Uses

- Real-time display at video rates
- Enable image pan, zoom, scroll

Location

Usually part of the image hardware unit.

Image Displays

Typical Devices

- Flat-panel monitors (color)
- Driven by standard display cards

Advanced Display

- Stereo viewing via head-mounted displays

Hardcopy Devices

Examples

- Laser printers
- Inkjet printers
- Film writers
- CD/DVD writers

Usage

- Paper: written documents
- Film: high-resolution output
- Presentations: digital projection or transparencies

Networking

Importance

Vital for image transmission, especially in multi-system environments.

Key Concern

Bandwidth—large image files require high-speed data transfer.

- Optical fiber and broadband improving efficiency.
- Remote site transfers may still be slower.

Summary

Core System Components

- ① Image Sensing (Sensor + Digitizer)
- ② Specialized Hardware (e.g., ALU, Frame Buffers)
- ③ Computer (Processing + Control)
- ④ Software (Modular, Programmable)
- ⑤ Mass Storage (RAM, Disk, Archival)
- ⑥ Display and Hardcopy Devices
- ⑦ Networking (Bandwidth, Connectivity)

Trend

More compact, software-driven, and accessible systems.